One cosine cannot read what a translation keeps

Translation fidelity lives on the directions of the embedding space a reader cares about. One cosine flattens every direction into a single number and privileges none. Which directions are salient is an open question that STRATA does not answer, so this note takes two that are obviously salient, colour and sound, chosen by hand. In a constructed four-cell demonstration a filler word's colour reading drops by two dictionary standard deviations while its cosine to the source stays above 99.9 percent of the dictionary. In the STRATA census translators keep colour in 88 percent of the 129 crossings where the source line carries it, and seven renderings of seven recreate a sound the probe cannot find in the source line.

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A fidelity metric built on a multilingual encoder scores a translation with one number, the cosine between source and translation embeddings (Feng et al. 2022). It says how far apart two texts sit, not what moved. This note argues that fidelity lives on a few directions of that space, the ones a reader registers, and that the cosine cannot single them out. A constructed demonstration on one line shows the geometry, that a colour axis can move far while the cosine barely moves. The STRATA census of 4,143 real crossings shows translators keeping colour where the source has it and recreating sound where the source wording never states it.

WHAT THE COSINE CANNOT SINGLE OUT

An encoder maps a text to a unit vector of 768 numbers. Any direction in that space is an axis, and a text's reading on an axis is the projection of its vector, after a whitening step, onto that direction. STRATA fixes a few axes in advance, among them colour and sound. A word in a line fires the colour detector when deleting it drops the line's colour reading past a cut fixed before any run. A line where the detector fires but no published inventory claims a word is recorded as a ghost, colour attested but unclaimed.

The cosine is the single scalar a plain fidelity metric sees, the dot product of two unit vectors, whether of two lines or of two words embedded alone. It sums agreement over all 768 directions at once and privileges none. It cannot tell a reader-salient, meaning-bearing axis from any other direction, so a rendering that lost the colour can score the same as one that lost nothing. That indifference is the limitation this note attacks, and it is a limitation of the number, not of the model. The colour is in the embedding. The cosine has no way to ask for it.

Which axes a metric should privilege is a prior question, and STRATA does not answer it. It has no way of its own to find the salient axes. The one signal of human salience in hand is a reader study, four readers tagging nine poems line by line in their own words, free to coin any field. Colour and sound were among the fields that surfaced and survived consolidation. So this note takes two axes that are trivially salient to a reader, chosen by hand. Both axes rest on datasets that record human attention to exactly these attributes. The colour axis is built from COCO captions, people describing photographs, with the colour word checked against the image's measured chroma (Lin et al. 2014). The sound axis is built from AudioCaps captions of AudioSet clips, people describing what they hear (Gemmeke et al. 2017). Nothing below claims that STRATA discovers which axes matter. It claims that on two axes which plainly do, translators behave in a way one cosine cannot see.

An implicit-colour word is not a colour term but carries strong colour attributes, French *chaude*, warm, or *earth*. No colour inventory claims them, yet a reader sees a warm light as golden and the earth as brown. The cosine is blindest to implicit colour, because nothing on the word's surface marks the colour the reader supplies. That blindness makes such a word the sharp vehicle, where a plain colour term would blunt it.

A QUALITATIVE DEMONSTRATION, NOT A BENCHMARK

The vehicle is *chaude* in Baudelaire's *Dans une chaude lumière.* There *chaude* fires the colour detector, at a shift of 0.027 against a cut of 0.015, and no French colour inventory claims it, so the line reads a ghost. The translation blank is *In a ____ light.*, filled four times by construction.

The grid has two dimensions with two levels each, measured against the source word embedded alone. On the colour axis a filler is near if its reading matches the source word's and far if it sits at the fifth percentile of a dictionary of 3,339 common English adjectives, two and a half dictionary standard deviations (SD) lower. On the cosine a filler is near if it matches the closest any dictionary word comes to the source, 0.967, and far if it matches the dictionary median, 0.452. For each cell I build a point at its two target coordinates, the colour reading and the cosine, then decode the nearest natural word to that point, by a rule that reads both the embedding and the colour coordinate. Because the point is built and not searched for, all four cells fill. The table gives each decoded word's measured coordinates, not the target it was aimed at.

cell	filler	cosine to *chaude* (word, dictionary rank)	colour reading (word, distance from source)	translation	line cosine to the source line	line colour reading (z against news sentences)	colour verdict against the source line
axis near, cosine near	hot	0.967, the closest word	+0.012, 0.2 SD below	In a hot light.	0.902	+0.03	ghost kept
axis far, cosine near	heated	0.867, above 99.9 %	−0.033, 2.3 SD below	In a heated light.	0.854	−0.56	ghost lost
axis near, cosine far	abiding	0.417, 29th percentile	+0.015, 0.1 SD below	In an abiding light.	0.698	+0.30	ghost kept
axis far, cosine far	embodied	0.435, 39th percentile	−0.039, 2.5 SD below	In an embodied light.	0.687	−0.96	ghost lost
reference, not a cell	warm	0.965	+0.003, 0.7 SD below	In a warm light.	0.911	+0.08	ghost kept

The last column is the verdict of the census comparator, which crosses the source line's colour state against the rendering's and here reads whether the ghost survived. The axis-near lines keep the ghost, the axis-far lines lose it, and the cosine-near lines sit above 99.9 percent of the 3,339 dictionary lines in cosine to the source line. A decode by cosine alone cannot even fill the far cells. It returns *hot* for both cosine-near cells, because dropping the colour reading by two and a half standard deviations displaces the vector by only 0.01 in cosine.

The prediction is about magnitude. Read down the table and a reader should register a large change in meaning along the colour dimension and a small but real change along the cosine dimension. The cosine is not null, but the typed axis carries far more of what a reader registers as meaning change, and for implicit colour most of it. Quality is the reader's call, and no model and no author has ranked the four translations. This is a qualitative demonstration on one line, one encoder and one dictionary, not a benchmark, and it makes no accuracy claim.

FIRST PROOF FROM THE PILOT, TRANSLATORS KEEP COLOUR

The census scores 64 translation pairs of nine poems line by line, 4,143 crossings of a source line against its rendering on one axis. On colour, the axis validated at the grain of a line, the source line carries the colour in 129 crossings, in a word a published inventory claims or a detector fire where that language has no inventory.

census of record, the source line carries the axis	crossings	share
colour (n = 129), rendering still carries it in some form (stated, carried by form or referent, or detector-attested)	113	88 %
colour, rendering states it in a word an inventory claims	72	56 %
colour, rendering lost it outright	16	12 %
sound (n = 536), still carried in some form	422	79 %
all five STRATA axes (n = 2,088), still carried in some form	1,635	78 %

Read on wording alone, a translator keeps the colour in just over half the crossings. Read on the typed axis, in nearly nine of ten. The crossings that move between the two readings are ones where the rendering dropped the inventory word while the detector still fires on the line. Keeping is the translator's vote on what matters, and an axis kept in nine crossings of ten is one a metric has to be able to read.

SECOND PROOF FROM THE PILOT, TRANSLATORS RECREATE SOUND THE PROBE CANNOT SEE

The fourth line of the tenth of the Nineteen Old Poems reads 札札弄机杼, the loom's clatter in onomatopoeia. No Chinese inventory claims the word, and no word moves the sound detector past its cut, the largest shift being 0.017 against a cut of 0.024, so the source line is silent to the instrument. Seven renderings sit on that line, six by human translators in English and German, the dated ones from 1899 to 1996, and one machine control. All seven bring the sound back. Six state it in a word the English sound inventory claims, clack, clack, click, clack, tune and clatter. The seventh, Forke's German *Webstuhl klappert laut*, fires the detector hardest of all, at 0.151, where no inventory of ours runs. The instrument corroborates the convergence on the rendering side, six receipts and one detector fire, and not on the source side. The pattern recurs on colour, where line 15 of *L'Albatros* states no colour and fires no colour token, yet five English translators independently render *sol* as *earth*.

A probe that cannot find the axis in the source, and seven translators who all produce it, together show that the axis

is real to readers and that the probe is blind to it there. A cosine between the Chinese line and any of the seven has no slot for a sound that was in the line and was written down seven times. A typed slot does.

CONCLUSION

Translators preserve the salient axes, and sometimes invent them. Colour survives in nine crossings of ten and sound returns in seven renderings of seven where the source wording never states it. A fidelity metric should therefore read these typed directions, and the cosine cannot, because it flattens every direction and singles out none. The prior question stays open and is named here, not hedged. Which axes are salient is something STRATA cannot yet identify. Four readers and two hand-picked axes are the whole of the evidence on salience in hand, and finding the salient axes from readers, from data, or from the encoder′s own geometry is the question to pursue next. How a typed metric should weight the axes it reads is a second, starting with ratings of the four translations above. The question a reader asks of a translation was never how far it moved. It was what of the line is still there.

Feng, F., Yang, Y., Cer, D., Arivazhagan, N., Wang, W. (2022). Language-Agnostic BERT Sentence Embedding. *ACL 2022*, 878–891. · Lin, T.-Y. et al. (2014). Microsoft COCO. *ECCV 2014*. · Gemmeke, J. F. et al. (2017). Audio Set. *ICASSP 2017*. · Census and instrument, Lu (2026), *STRATA: Discovering a Text′s Schema and Measuring Its Survival in Poetic Translation*. Census counts from `findings_v51_linegrain_0728_62.json`, loom facts from `albatros_L15_exemplar_gen_63.py`, axes `color_salience_axis_48.npz` and `sound_salience_axis_v3_49.npz`, reader study `design/marking_pack_v2.md`. Demonstration code and per-cell numbers in `twobytwo.py` and `results/twobytwo_results.json`.